Supplementary Material: Mapping Verbal Survey Responses to Numerical Acceptance Probabilities

The survey measures riders' stated acceptance intentions using five verbal response categories: *"definitely cancel"*, *"very likely cancel"*, *"uncertain"*, *"very likely accept"*, and *"definitely accept"*. To convert these labels into numerical probabilities suitable for estimation, we adopt a calibrated mapping procedure consisting of two steps.

First, following the *IPCC calibrated language scale* (Mastrandrea et al., 2010), we assign the five verbal categories to initial objective probabilities

$$\{0.00,\ 0.10,\ 0.50,\ 0.90,\ 1.00\},$$

respectively. This provides an interpretable ordinal anchor for each response level.

Second, to better capture the nonlinear and asymmetric perception of probability in human judgment, we transform these objective anchors using the cumulative prospect theory (CPT) weighting function proposed by Tversky and Kahneman (1992):

$$w(p) = \frac{p^\gamma}{\left(p^\gamma + (1-p)^\gamma\right)^{1/\gamma}}, \tag{1}$$

where $\gamma = 0.61$ follows the empirical evidence reported in Gonzalez and Wu (1999). This transformation preserves the ordinal ranking of the response categories while allowing for psychologically more realistic spacing across adjacent levels.

Applying this function to the intermediate anchors yields the final encoded probabilities

$$\{0.00,\ 0.14,\ 0.50,\ 0.93,\ 1.00\},$$

corresponding to *"definitely cancel"*, *"very likely cancel"*, *"uncertain"*, *"very likely accept"*, and *"definitely accept"*, respectively. These encoded values are then used as the dependent variable in the regression analysis for estimating the acceptance probability function.

Participants were informed of the platform's baseline pricing rule—based on trip distance and travel duration—before providing their responses, so that the elicited acceptance intentions reflect behavior under known price conditions.

References

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